Tim Tomov

PhD Student in Computer Science · Reliable AI, Large Language Models, Uncertainty Quantification

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EDUCATION

Ph.D. in Computer Science Nov 2025 – expected 10/29
Technical University of Munich · Advisor: Prof. Stephan Günnemann *Munich, DE*

M.Sc. Mathematics in Data Science (Grade: 1.3; Thesis: 1.0) Oct 2022 – Sep 2025
Technical University of Munich *Munich, DE*

B.Sc. Information Systems (Grade: 1.5) Oct 2017 – Jul 2021
Technical University of Munich *Munich, DE*

SELECTED PUBLICATIONS

1. Task-Awareness Improves LLM Generations and Uncertainty.
   Tim Tomov*, Dominik Fuchsgruber*, Stephan Günnemann
   International Conference on Machine Learning (ICML), 2026. arXiv

2. What Is a Clarification? Estimating Model-Dependent Input Ambiguity.
   Jingqin Tang*, **Tim Tomov***, Sebastian Schmidt, Dominik Fuchsgruber, Stephan Günnemann
   LP4FM @ NeurIPS 2026 (Oral). **Outstanding Paper Award.** OpenReview

3. Task-Aware Calibration: Provably Optimal Decoding in LLMs.
   Tim Tomov*, Dominik Fuchsgruber*, Rajeev Verma, Stephan Günnemann
   EIML @ ICML 2026 (Spotlight); Agentic UQ @ ICML 2026. arXiv

4. The Illusion of Certainty: Uncertainty Quantification for LLMs Fails under Ambiguity.
   Tim Tomov, Dominik Fuchsgruber, Tom Wollschläger, Stephan Günnemann
   arXiv preprint, 2025. Preliminary version: *EIML @ EurIPS 2025 (Oral); SPIGM @ NeurIPS 2025.* arXiv

* Equal contribution

RESEARCH EXPERIENCE

PhD Researcher Nov 2025 – present
Data Analytics and Machine Learning Group, Technical University of Munich *Munich, DE*

- Research agenda: uncertainty quantification & decision-theoretic foundations for reliable LLMs
- In first year: First author of an ICML 2026 main-track paper [1], a NeurIPS workshop Oral with Outstanding Paper Award [2], and an ICML workshop Spotlight [3]
- Run LLM experiments with PyTorch, Hugging Face Transformers, and SLURM-managed GPU clusters

Master's Thesis Research Oct 2024 – Sep 2025
Data Analytics and Machine Learning Group, Technical University of Munich *Munich, DE*

- Showed that current LLM uncertainty methods fail to estimate epistemic uncertainty once aleatoric uncertainty is present; thesis *Epistemic Uncertainty in Large Language Models* led to [4]

Research Scientist Intern Apr 2024 – Jul 2024
Amazon *Munich, DE*

- Developed post-hoc threshold optimization for asymmetric-cost classification using genetic algorithms
- Benchmarked LLMs for large-scale, highly imbalanced classification under asymmetric costs

Research Project Oct 2023 – Mar 2024
Visual Computing Group, Technical University of Munich *Munich, DE*

- Conditioned Stable Diffusion on depth, semantics, and pose for single-image novel-view synthesis

Research Assistant May 2021 – Nov 2021
Image-Based Biomedical Modeling Group, Technical University of Munich *Munich, DE*

- Extended 3D U-Net architectures with attention mechanisms for MRI-based sarcoma segmentation

TALKS & PRESENTATIONS

ICML 2026 Two poster presentations [1, 3]; spotlight talk, EIML Workshop [3]

NeurIPS 2026 Oral, LP4FM Workshop (upcoming) [2]

EurIPS 2025 Oral, EIML Workshop: preliminary version of *The Illusion of Certainty* [4]

REVIEWING

ICLR 2027 Reviewer

TECHNICAL SKILLS

Research LLMs, Uncertainty Quantification, Decision Making

Machine Learning PyTorch, Hugging Face Transformers, Lightning, Diffusers, Scikit-learn, NumPy

Programming & Systems Python, SQL, R, SLURM, Git, Amazon SageMaker

ADDITIONAL INDUSTRY EXPERIENCE

Associate Data Scientist Jan 2023 – Nov 2023
Convalid Analytics · Working Student *Munich, DE*

- Built topic-modeling and summarization pipelines using spherical k-means, OPTICS, and LLMs
- Built a RAG question-answering system over 100k+ customer feedback entries
- Applied SetFit for efficient few-shot classification of customer complaints

AWARDS & AFFILIATIONS

PhD Scholarship Recipient Konrad Zuse School of Excellence in Reliable AI (relAI)

Junior Member Munich Center for Machine Learning (MCML)

Erasmus+ Scholarship M.Sc. Data Science exchange, Newcastle University